JAVASCRIPT | ASSIGNMENT 3

Conditional Logic Implementation

Task: Create a script that evaluates a user's numerical test score using if...else statements, determines a letter grade using a switch statement, and uses a ternary operator to print whether they passed or failed.

INPUT

```
score = 85
```

CODE

```
// Assignment 3
let score = 85;

// if...else: validate the score
if (score < 0 || score > 100) {
  console.log("Invalid score");
} else {
  // switch: determine the letter grade
  let grade;
  switch (true) {
    case score >= 90:
      grade = "A";
      break;
    case score >= 80:
      grade = "B";
      break;
    case score >= 70:
      grade = "C";
      break;
    case score >= 60:
      grade = "D";
      break;
    default:
      grade = "F";
  }

  // ternary: pass or fail
  let status = score >= 60 ? "Passed" : "Failed";

  console.log("Score: " + score);
  console.log("Grade: " + grade);
  console.log("Status: " + status);
}
```

OUTPUT

```
Score: 85
Grade: B
Status: Passed
```

NOTES

- switch (true) lets each case hold a condition, which suits score ranges.
- Each break stops execution from falling through to the next case.
- The ternary condition ? a : b is a one-line if...else.